**B.Tech. 8th Semester Examination, 2024**

*Course: B.Tech.* *Time: 03 Hours*
*Code: 105816* *Subject: Embedded Systems* *Full Marks: 70*

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

***Q.1 Choose the correct answer of the following (any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following task swapping methods is a better choice in the embedded system design?
(i) Time slicing technique. (ii) Rate monotonic scheduling technique.
(iii) Cooperative multitasking technique. (iv) Preemption technique.

(b) Which of the following statements is TRUE for the Von Neumann architecture?
(i) Separate bus between program (or code) memory and data memory.
(ii) Shared bus between program (or code) memory and data memory.
(iii) External bus for program (or code) memory and data memory.
(iv) External bus for data memory only.

(c) Which of the following microprocessors is/are designed by Zilog, Inc. for use in embedded systems?
(i) Z80. (ii) 80386.
(iii) 8087. (iv) All of the given options.

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(d) ARM instruction set architecture is divided into _____ classes of instructions.
(i) Two. (ii) Four.
(iii) Six. (iv) Eight.

(e) What does CPI stand for?
(i) Complex Program Instruction. (ii) Cycles per Instruction.
(iii) Current Program Instruction. (iv) None of the given options.

(f) Which of the following is the biggest challenge in cache memory design?
(i) Coherency. (ii) Memory access.
(iii) Memory capacity. (iv) None of the given options.

(g) Which memory component(s) of memory hierarchy is/are typically located on embedded board, but outside of master processor?
(i) Level-2 Cache. (ii) Main Memory.
(iii) Secondary Memory. (iv) All of the given options.

(h) Is UART (Universal Asynchronous Receiver-Transmitter) an example of a synchronous serial interface?
(i) TRUE. (ii) FALSE.

(i) What component(s) on an embedded board interconnect different buses for carrying information from one bus to another?
(i) CDROM Drive. (ii) MMU.
(iii) Bridge. (iv) All of the given options.

(j) Which of the following options is/are the common method(s) for connecting a peripheral to a processor?
(i) Software. (ii) External interrupts.
(iii) Internal interrupts. (iv) All of the given options.



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*Q.2* (a) What are the basic materials in which all components on an embedded board are comprised of? Give two examples of each type of material. **[6]**

(b) According to the Von-Neumann architecture, list and define the different components of the CPU in an embedded system. **[8]**

*Q.3* (a) Describe the memory hierarchy of an embedded system, with illustrations. **[10]**

(b) Write down the differences between MMUs and memory controllers in an embedded system. **[4]**

*Q.4* (a) What are the differences between serial I/O and parallel I/O of an embedded system board? **[3]**

(b) Describe how the synchronous transfer of serial data works between the embedded system transmitter and receiver, with illustrations. **[6]**

(c) Draw and describe the schematic diagram I/O subsystem of an embedded system. **[5]**

*Q.5* (a) What is a bus in an embedded system board? What are its purposes? **[4]**

(b) Describe the categories of board buses of embedded system. **[7]**

(c) What are the differences between expandable bus and non-expandable bus? **[3]**

*Q.6* (a) Describe any three bus arbitration schemes. **[6]**

(b) What is bus handshaking? What is the basis of a bus handshake? **[5]**

(c) What is the difference between bus bandwidth and bus width? **[3]**

*Q.7* (a) Describe the different phases of an embedded system design and development lifecycle model, with illustrations. **[10]**

(b) What are the differences between testing, validation and verification in the design process of an embedded system? **[4]**

*Q.8* (a) Define the following metrics for measurements of CPU's performance: **[10]**

(i) Execution time
(ii) Throughput
(iii) Predictability
(iv) Instruction issue width
(v) Dynamic voltage and frequency scaling (DVFS)

(b) Name the stages of an ARM pipeline. **[4]**

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*Q.9* Write short notes on ***any two*** of the following: **[7x2=14]**

(a) Super Harvard Architecture Single-Chip Computer (SHARC).
(b) Static and dynamic power management of CPU.
(c) Cache memory in embedded system.
(d) Assembler and linker.



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